

# ADI-FDTD for Inhomogeneous Media

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## 1 Preliminaries

### 1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \quad (1.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (1.1b)$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \epsilon \frac{\partial E_x}{\partial t} + \sigma E_x \quad (1.2a)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \epsilon \frac{\partial E_y}{\partial t} + \sigma E_y \quad (1.2b)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \epsilon \frac{\partial E_z}{\partial t} + \sigma E_z \quad (1.2c)$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \left( \frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \right) \quad (1.3a)$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu} \left( \frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \right) \quad (1.3b)$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu} \left( \frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \right) \quad (1.3c)$$

### 1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at  $i + \frac{1}{2}$ , while a nodal direction has index type 1 and is located at  $i$ . The implementation uses

$\mathbf{E}_\alpha$  : cell-centered in the  $\alpha$  direction and nodal in the other two directions,

$\mathbf{H}_\alpha$  : nodal in the  $\alpha$  direction and cell-centered in the other two directions.

Thus the component locations are

| Component     | $x$ location      | $y$ location      | $z$ location      |
|---------------|-------------------|-------------------|-------------------|
| $E_{x,i,j,k}$ | $i + \frac{1}{2}$ | $j$               | $k$               |
| $E_{y,i,j,k}$ | $i$               | $j + \frac{1}{2}$ | $k$               |
| $E_{z,i,j,k}$ | $i$               | $j$               | $k + \frac{1}{2}$ |
| $H_{x,i,j,k}$ | $i$               | $j + \frac{1}{2}$ | $k + \frac{1}{2}$ |
| $H_{y,i,j,k}$ | $i + \frac{1}{2}$ | $j$               | $k + \frac{1}{2}$ |
| $H_{z,i,j,k}$ | $i + \frac{1}{2}$ | $j + \frac{1}{2}$ | $k$               |

The half-cell offsets therefore do not appear explicitly as array indices such as  $i + \frac{1}{2}$ . They are encoded in the **MultiFab** staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example,  $H_{x,i,j,k}$  is located at  $j + \frac{1}{2}$  in the  $y$  direction, while  $E_{z,i,j,k}$  and  $E_{z,i,j+1,k}$  are located at  $j$  and  $j + 1$ . Therefore

$$\left. \frac{\partial E_z}{\partial y} \right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}.$$

Similarly,  $E_{x,i,j,k}$  is located at  $j$ , while  $H_{z,i,j-1,k}$  and  $H_{z,i,j,k}$  are located at  $j - \frac{1}{2}$  and  $j + \frac{1}{2}$ , so

$$\left. \frac{\partial H_z}{\partial y} \right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}.$$

This is why the magnetic update below uses forward differences such as  $j + 1$  and  $k + 1$ , while the electric update uses backward differences such as  $j - 1$  and  $k - 1$ : each derivative is centered at the staggered location of the field component being updated.

## 2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update  $\mathbf{H}$  by a half step, then update  $\mathbf{E}$  by a full step, and finally update  $\mathbf{H}$  by another half step. Define the update coefficients

$$C_a = \frac{2\epsilon - \sigma\Delta t}{2\epsilon + \sigma\Delta t}, \quad C_b = \frac{2\Delta t}{2\epsilon + \sigma\Delta t}, \quad D_b = \frac{\Delta t}{\mu}, \quad (2.1)$$

where  $\Delta t$  is the time step,  $\epsilon$  is the permittivity,  $\mu$  is the permeability, and  $\sigma$  is the conductivity.

### 2.1 First Magnetic Half-Step

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{2} \left[ \frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} - \frac{E_{y,i,j+\frac{1}{2},k+1}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta z} \right] \quad (2.2a)$$

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - \frac{D_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{2} \left[ \frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} - \frac{E_{z,i+1,j,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta x} \right] \quad (2.2b)$$

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - \frac{D_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{2} \left[ \frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j+1,k}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta y} \right]. \quad (2.2c)$$

In the AMReX Yee grid, with homogeneous media, the magnetic field updates are implemented as

$$B_x(i, j, k) = \frac{1}{2\Delta t} \left[ \frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.3a)$$

$$B_y(i, j, k) = \frac{1}{2\Delta t} \left[ \frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.3b)$$

$$B_z(i, j, k) = \frac{1}{2\Delta t} \left[ \frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.3c)$$

### 2.2 Electric Full-Step

$$E_{x,i+\frac{1}{2},j,k}^{n+1} = C_{a,i+\frac{1}{2},j,k}^{n+1} E_{x,i+\frac{1}{2},j,k}^n + C_{b,i+\frac{1}{2},j,k}^{n+1} \left[ \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right] \quad (2.4a)$$

$$E_{y,i,j+\frac{1}{2},k}^{n+1} = C_{a,i,j+\frac{1}{2},k}^{n+1} E_{y,i,j+\frac{1}{2},k}^n + C_{b,i,j+\frac{1}{2},k}^{n+1} \left[ \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x} \right] \quad (2.4b)$$

$$E_{z,i,j,k+\frac{1}{2}}^{n+1} = C_{a,i,j,k+\frac{1}{2}}^{n+1} E_{z,i,j,k+\frac{1}{2}}^n + C_{b,i,j,k+\frac{1}{2}}^{n+1} \left[ \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right]. \quad (2.4c)$$

In the AMReX Yee grid, with homogeneous media, the electric field updates are implemented as

$$E_x(i, j, k) + = c^2 \Delta t \left[ \frac{B_z(i, j, k) - B_z(i, j - 1, k)}{\Delta y} - \frac{B_y(i, j, k) - B_y(i, j, k - 1)}{\Delta z} \right] \quad (2.5a)$$

$$E_y(i, j, k) + = c^2 \Delta t \left[ \frac{B_x(i, j, k) - B_x(i, j, k - 1)}{\Delta z} - \frac{B_z(i, j, k) - B_z(i - 1, j, k)}{\Delta x} \right] \quad (2.5b)$$

$$E_z(i, j, k) + = c^2 \Delta t \left[ \frac{B_y(i, j, k) - B_y(i - 1, j, k)}{\Delta x} - \frac{B_x(i, j, k) - B_x(i, j - 1, k)}{\Delta y} \right]. \quad (2.5c)$$

### 2.3 Second Magnetic Half-Step

$$H_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[ \frac{E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right] \quad (2.6a)$$

$$H_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[ \frac{E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} \right] \quad (2.6b)$$

$$H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[ \frac{E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right]. \quad (2.6c)$$

In the AMReX Yee grid, the second magnetic field updates are implemented as

$$B_x(i, j, k) - = \frac{1}{2\Delta t} \left[ \frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.7a)$$

$$B_y(i, j, k) - = \frac{1}{2\Delta t} \left[ \frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.7b)$$

$$B_z(i, j, k) - = \frac{1}{2\Delta t} \left[ \frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.7c)$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \leq \frac{1}{c \sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}}}, \quad c = \frac{1}{\sqrt{\mu \epsilon}}. \quad (2.8)$$

## 3 ADI Method

The ADI-FDTD update advances the fields by a full time step  $\Delta t$  in two half steps of size  $\Delta t/2$ . Unlike the explicit three-stage leapfrog update above, each ADI half step first updates  $\mathbf{E}$  implicitly over  $\Delta t/2$ , then updates  $\mathbf{H}$  explicitly over  $\Delta t/2$  from the updated electric field. The first half step treats one set of implicit directions; the second half step treats the complementary directions to complete the cycle. Define the half-step update coefficients

$$C_a = \frac{4\epsilon - \sigma \Delta t}{4\epsilon + \sigma \Delta t}, \quad C_b = \frac{2\Delta t}{4\epsilon + \sigma \Delta t}, \quad D_b = \frac{\Delta t}{2\mu}, \quad (3.1)$$

where  $\Delta t$  is the time step,  $\epsilon$  is the permittivity,  $\mu$  is the permeability, and  $\sigma$  is the conductivity. These differ from the full-step coefficients in (2.1) because each electric update uses  $\Delta t/2$ .

### 3.1 First ADI Half-Step

By equation (1.2a),

$$\epsilon_{i+\frac{1}{2}, j, k} \frac{E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} - E_{x, i+\frac{1}{2}, j, k}^n}{\Delta t/2} + \sigma_{i+\frac{1}{2}, j, k} \frac{E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} + E_{x, i+\frac{1}{2}, j, k}^n}{2} = [H_1], \quad (3.2)$$

where

$$[H_1] = \frac{H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} - H_{z, i+\frac{1}{2}, j-\frac{1}{2}, k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^n - H_{y, i+\frac{1}{2}, j, k-\frac{1}{2}}^n}{\Delta z}. \quad (3.3)$$

Rearranging gives

$$\left(\frac{2\epsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} = \left(\frac{2\epsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^n + [H_1]. \quad (3.4)$$

Hence

$$E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^n + C_{b,i+\frac{1}{2},j,k} [H_1], \quad (3.5)$$

By equation (1.3c),

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left( \frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} \right), \quad (3.6)$$

Substituting the magnetic update into (3.5) and eliminate  $H_z^{n+\frac{1}{2}}$  terms gives

$$\begin{aligned} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} &= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^n \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left[ H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left( \frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} \right) \right. \\ &\quad \left. - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n - D_{b,i+\frac{1}{2},j-\frac{1}{2},k} \left( \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n}{\Delta x} \right) \right] \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left( H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right). \end{aligned} \quad (3.7)$$

Collecting the unknown  $E_x^{n+\frac{1}{2}}$  terms gives

$$\begin{aligned} &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} \\ &+ \left[ 1 - \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} \right] E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} \\ &= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^n + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left( H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y \Delta x} \left( E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\ &+ \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y \Delta x} \left( E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left( H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right). \end{aligned} \quad (3.8)$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{x1}E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1}E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1}E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} &= p_{x1}E_{x,i+\frac{1}{2},j,k}^n \\
&+ \frac{1}{\Delta y} \left( H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\
&- \frac{1}{\Delta z} \left( H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\
&+ \frac{q_{x1}}{\Delta x \Delta y} \left( E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\
&- \frac{r_{x1}}{\Delta x \Delta y} \left( E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right). \tag{3.9}
\end{aligned}$$

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{3.10}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{3.11}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{3.12}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.13}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{3.14}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.15}$$